

Hirschberg's Algorithm

MPA-PRG: Programming in Bioinformatics

Exercise 5

Sequence Alignment

- inputs:
 - two sequences ($X = x_1x_2 \dots x_m$ and $Y = y_1y_2 \dots y_n$)
 - scoring system – match/mismatch scores and gap penalty
- global alignment – Needleman-Wunsch algorithm
- local alignment – Smith-Waterman algorithm
- time complexity: $O(mn)$
- space complexity: $O(mn)$

Alignment between Prefixes of X and Y

		Y																				
	S	j	A	A	G	G	T	A	T	G	A	A	T	C								
	i	0	1	2	3	4	5	6	7	8	9	10	11	12								
X	A	1																				
	A	2																				
	C	3																				
	G	4																				
	T	5																				
	T	6																				
	G	7																				
	A	8																				

the best alignment between
prefix $x_1 \dots x_4$ and prefix $y_1 \dots y_4$

Linear Space Alignment

		Y												
S		j	A	A	G	G	T	A	T	G	A	A	T	C
i		0	1	2	3	4	5	6	7	8	9	10	11	12
X	A	1												
	A	2												
	C	3												
	G	4												
	T	5												
	T	6												
	G	7												
	A	8												

we need only three values:

- $S(i - 1, j)$
- $S(i, j - 1)$
- $S(i - 1, j - 1)$

therefore, we only need to keep previous and current row in memory

Alignment between Suffixes of X and Y

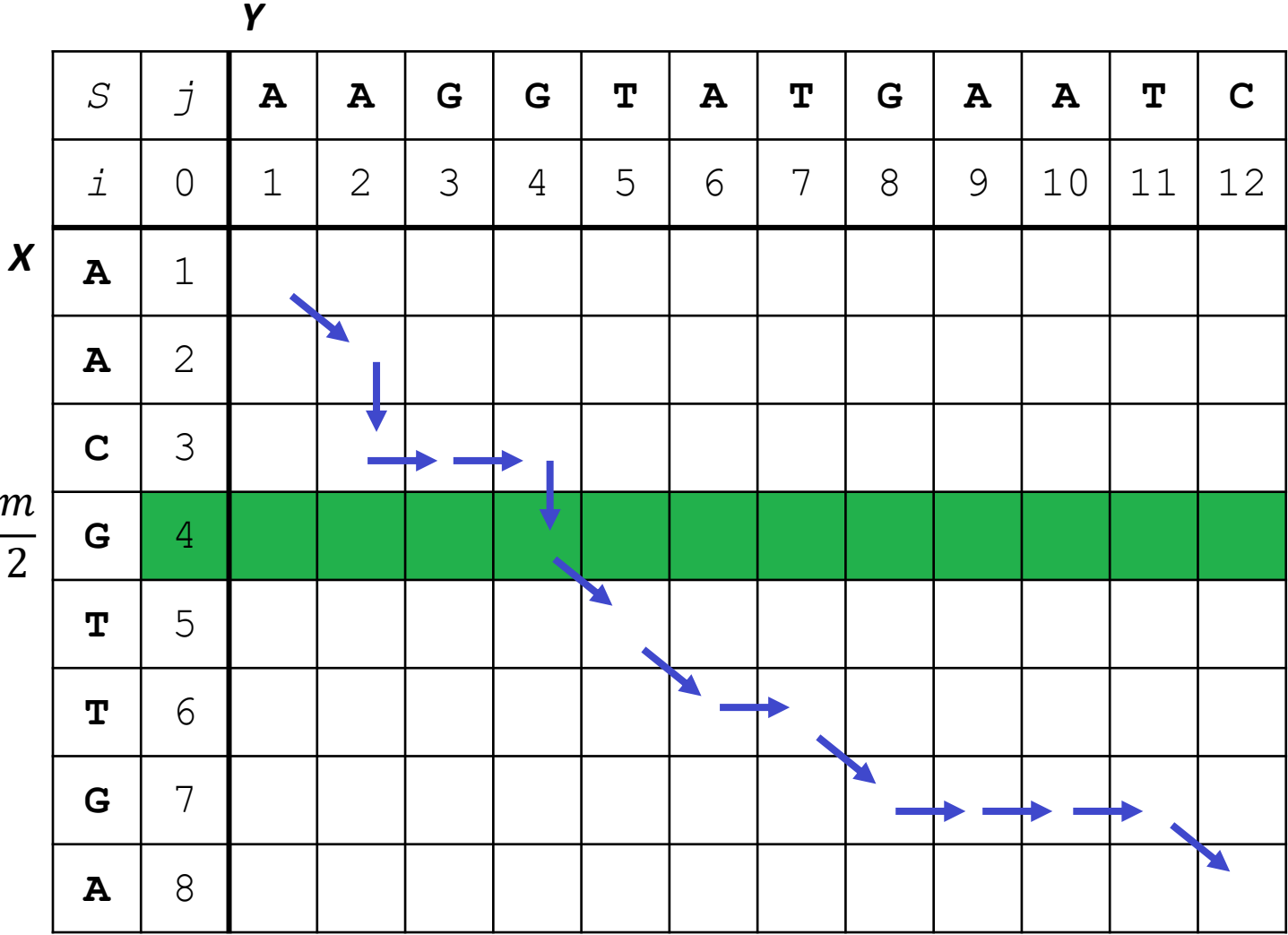
Y

<i>s</i>	<i>j</i>	A	A	G	G	T	A	T	G	A	A	T	C	
<i>i</i>	0	1	2	3	4	5	6	7	8	9	10	11	12	
X	A	1												8
	A	2												7
	C	3												6
	G	4												5
	T	5												4
	T	6												3
	G	7												2
	A	8												1
		12	11	10	9	8	7	6	5	4	3	2	1	0

the best alignment between
suffix $x_5 \dots x_8$ and suffix $y_5 \dots y_{12}$

the same principle as for prefix scoring

Path through Scoring Matrix



Path through Scoring Matrix

Y $j_{mid} = 4$

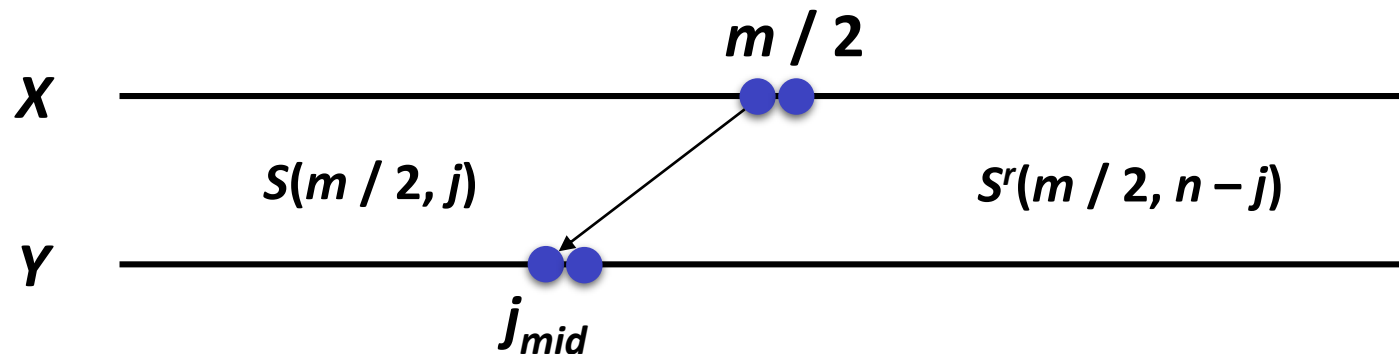
X	s	j	A	A	G	G	T	A	T	G	A	A	T	C
	i	0	1	2	3	4	5	6	7	8	9	10	11	12
$m/2$	A	1												
	A	2												
	C	3												
	G	4												
	T	5												
	T	6												
	G	7												
	A	8												

finding the actual alignment is equivalent to finding all cells that optimal path passes through

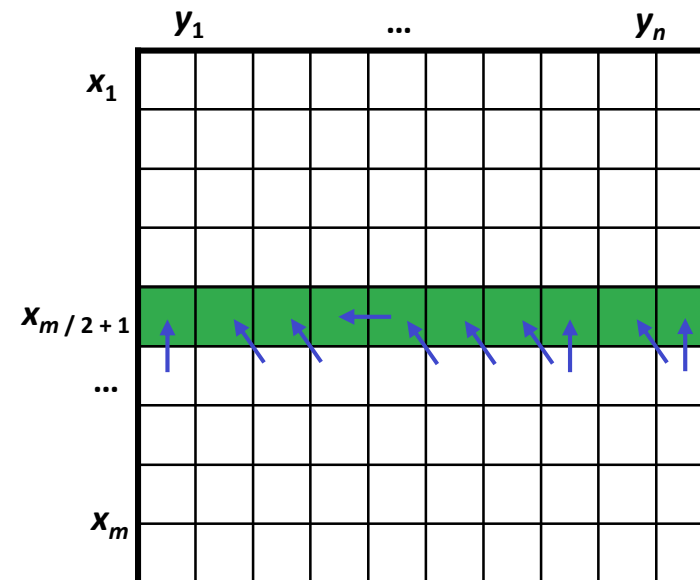
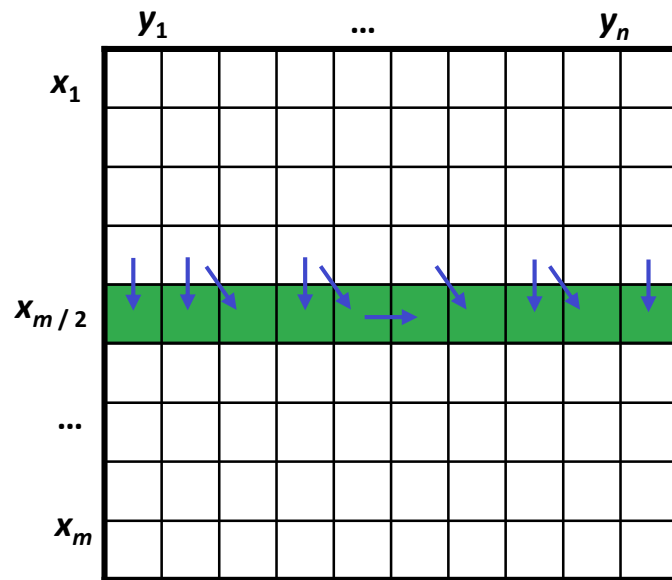
Hirschberg's Algorithm

- any part of matrix S from $(1,1)$ to (i,j) contains optimal path for subsequences $x_1x_2 \dots x_i$ and $y_1y_2 \dots y_j$
- X^r and Y^r are the reverse of X and Y , the optimal score (path) $S^r(i,j)$ of aligning $x_1^rx_2^r \dots x_i^r$ and $y_1^ry_2^r \dots y_i^r$ is the same as aligning $x_{m-i+1}x_{m-i+2} \dots x_m$ and $y_{n-j+1}y_{n-j+2} \dots y_n$
- assume m is even, then

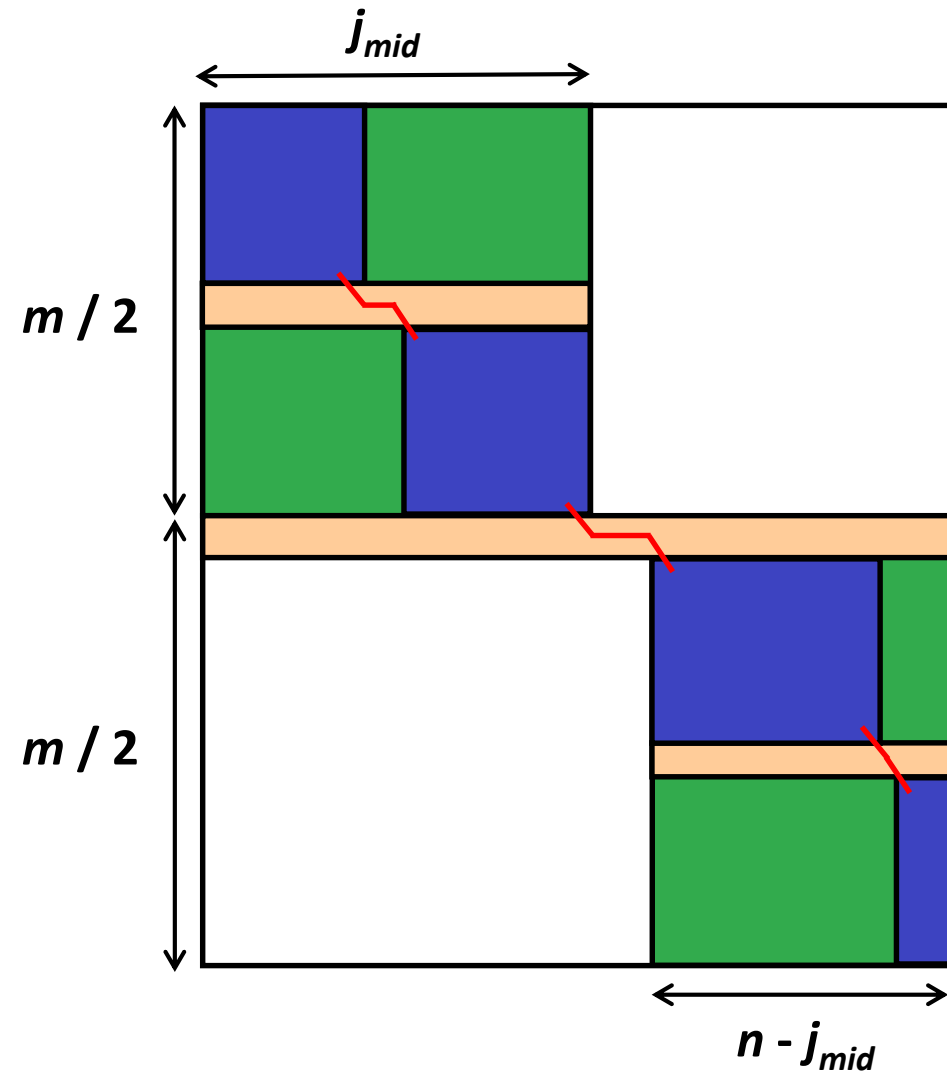
$$S(m,n) = \max_{j=0 \dots n} (S\left(\frac{m}{2}, j\right) + S^r\left(\frac{m}{2}, j - n\right))$$



Hirschberg's Algorithm



Hirschberg's Algorithm



Hirschberg's Algorithm

- for global alignment of two sequences
- uses recursion, divide and conquer and linear-space

Algorithm	Time Complexity	Space Complexity
Needleman-Wunsch	$O(mn)$	$O(mn)$
Hirschberg	$O(mn)$	$O(m + n)$

Example

- sequence 1: **AGTACGCA**
- sequence 2: **TATGC**
- match = 2
- mismatch = -1
- gap = -2

Example

- sequence 1: **TACGAGGCA**
- sequence 2: **ACGGA**
- match = 3
- mismatch = 1
- gap = -3

	A	C	G	G	A
0	-3	-6	-9	-12	-15
T	-3	1			
A	-6				
C	-9				
G	-12				
A	-15	-9	-3	3	7

	A	G	G	C	A
0	-3	-6	-9	-12	-15
A	-3				
G	-6				
G	-9				
G	-12	-6	0	6	8

1.	2.	3.	4.	5.	6.
-15	-9	-3	3	7	9
5	8	6	0	-6	-12
-10	-1	3	5	1	-3

$i_{mid} = 8 - 1 = 2$

find max

TACGA, AC		GGCA, GGA
A C		A C A
T		A
A		G -6 -2 2
C -9 -3 3		

G G A		A G G
G		A
G -6 0 6 3		C -6 0 4 1
-6 0 6 3		
1 4 0 -6		
-5 4 6 -3		

-3 -3 3		2 -2 -6		-7 -5 -3
2 -2 -6				
-7 -5 -3				

$i_{mid} = 3 - 1 = 2$

GA, --

G G, G G
G G
G -3 3 0
-3 3 0
-3 6 -3

G, G

G, G

-3 1		1		A -3 3
-3 1				
3 -3				
0 -2				

C, -

A, A

T A C G A G G C A		T A C, AC
- A C - - G G - A		A C C A
		C 3 3 0
		A -6 0 2
		-6 0 2
		0 3 -3
		-6 3 -1
		T A, A
		A A
		T -3 1
		A -3 3
		-3 1
		3 -3
		0 -2

C, C

T, -

A, A

Task

- in R, create a function `Hirschberg()` for the alignment of two sequences using Hirschberg's algorithm
- input:
 - DNASTring object representing NT or AA sequence to be aligned
 - DNASTring object representing NT or AA sequence to be aligned
 - a list of DNASTring objects with alignment of input sequences
 - an integer value of a score for matching bases
 - an integer value of a score for mismatching bases
 - an integer value of a penalty for gap insertion
- output:
 - a list of DNASTring objects with alignment of input sequences
- implement the function on your own or you can use a prepared template that is available on the GitHub page
- **hint:** you will need to implement your own function to calculate the score of the partial alignments (use the Needleman-Wunsch algorithm in linear space)